

# Data Mining (CS24303) 10–Page Master Revision

High–Yield Formulas, Schemas, Cuboid Lattices, Apriori Rules & Solved Numerical Cards

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# Data Mining & Data Warehousing (CS24303) 10–Page Master Quick Revision Guide

Formulas, Schemas, Cuboid Lattices, Apriori Rules, Cluster Algorithms & Solved Exam Cards

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## Master Formula & Metric Reference Sheet

| Domain                     | Universal Formula                                                                                                                                                   | Key Exam Property                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Min-Max Normalization      | $v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new\_max} - \text{new\_min}) + \text{new\_min}$                                                                     | Preserves exact relationships; sensitive to future out-of-bound outliers.         |
| Z-Score Standardization    | $z = \frac{v - \bar{x}}{\sigma}$                                                                                                                                    | Zero mean, unit variance ( $\mathcal{N}(0, 1)$ ); ideal when min/max are unknown. |
| $\chi^2$ Independence Test | $\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \quad \text{where } E_{ij} = \frac{\text{Row}_i \times \text{Col}_j}{N}$                                          | $df = (r - 1)(c - 1)$ ; tests categorical attribute independence.                 |
| Cosine Similarity          | $\text{sim}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\ \mathbf{x}\  \ \mathbf{y}\ } = \frac{\sum x_i y_i}{\sqrt{\sum x_i^2} \sqrt{\sum y_i^2}}$ | Angle between vectors; independent of document text length.                       |
| Jaccard Similarity         | $J(\mathbf{x}, \mathbf{y}) = \frac{q}{q + r + s}$                                                                                                                   | Asymmetric binary metric; completely ignores 0 – 0 negative matches $t$ .         |
| Association Rule Lift      | $\text{Lift}(A \implies B) = \frac{P(A \cup B)}{P(A)P(B)} = \frac{\text{Confidence}(A \implies B)}{\text{Support}(B)}$                                              | $\text{Lift} > 1 \implies$ Positive correlation; $= 1 \implies$ Independent.      |
| Information Gain           | $\text{Gain}(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{ S_v }{ S } H(S_v)$                                                                                 | Biased towards attributes with massive numbers of distinct values.                |
| Gini Impurity              | $\text{Gini}(S) = 1 - \sum_{i=1}^m p_i^2$                                                                                                                           | CART metric for strictly binary decision tree splits.                             |

## Complete 5–Module Comparative Checklists & Solved Flashcards

| Module                                     | Core Theoretical Concepts                                                                                              | High–Yield Exam Numericals                                                                               |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>M1: Data Understanding</b>              | KDD Process, Nominal/Ordinal/Interval/Ratio, Proximity Metrics (SMC, Jaccard, Cosine, Mahalanobis, Gower).             | Five-number summary, IQR outlier fences, Cosine vector similarity, Jaccard distance.                     |
| <b>M2: Preprocessing</b>                   | Missing imputation, Bin smoothing (means, boundaries), $\chi^2$ test, Min-Max / Z-score normalization, PCA.            | Contingency table $\chi^2$ computation, decimal scaling, Min-Max re-scaling to $[-1, 1]$ .               |
| <b>M3: Data Warehousing</b>                | OLTP vs OLAP, Star / Snowflake / Fact Constellation, Additive vs Semi-additive measures, Cuboid lattice combinatorics. | Total cuboids calculation with hierarchies $N = \prod (L_i + 1)$ , OLAP operations (slice/dice/roll-up). |
| <b>M4: Frequent Patterns</b>               | Apriori property, join & prune, FP-Tree conditional bases, ECLAT vertical TID lists, Closed vs Maximal.                | Full Apriori table trace, Support/Confidence/Lift calculations, FP-Tree branch insertion.                |
| <b>M5: Classification &amp; Clustering</b> | ID3 vs C4.5 vs CART, Naive Bayes, Confusion Matrix, ROC/AUC, k-Means vs PAM, DBSCAN density, LOF.                      | Entropy & Information Gain numerical, Confusion matrix Precision/Recall/F1, k-Means ID convergence.      |

### Solved Exam Flashcard 1: Star vs Snowflake Schema Differences

**Star Schema:** De-normalized dimension tables with redundant columns, single-join fact table queries, optimized for fast OLAP read speeds.

**Snowflake Schema:** Normalized dimension hierarchies (splitting out sub-dimensions into 3NF), zero redundancy, requires complex multi-table SQL joins that degrade interactive slice/dice query performance.

### Solved Exam Flashcard 2: k-Means vs DBSCAN Clustering Trade-offs

**k-Means:** Fast  $O(tkN)$ , requires upfront  $k$ , assumes spherical clusters of equal size/variance, highly sensitive to outliers.

**DBSCAN:** Density-based ( $\epsilon$ , MinPts), automatically determines number of clusters, discovers arbitrary non-convex geometries (crescents, rings), completely isolates noise outliers.

### Solved Exam Flashcard 3: Confidence vs Lift in Association Rules

Confidence measures conditional probability  $P(B \mid A)$ , but fails to account for the baseline popularity of item  $B$ . **Lift** divides confidence by  $P(B)$  to reveal true statistical correlation (**Lift**  $> 1$  positive correlation, **Lift**  $= 1$  independent, **Lift**  $< 1$  negative substitute).

### 🏛️ Solved Exam Flashcard 4: Confusion Matrix Metrics Summary

- Accuracy =  $\frac{TP+TN}{N}$  (Overall correctness)
- Sensitivity / Recall =  $\frac{TP}{TP+FN}$  (Fraction of actual positives captured)
- Specificity =  $\frac{TN}{TN+FP}$  (Fraction of actual negatives captured)
- Precision =  $\frac{TP}{TP+FP}$  (Fraction of predicted positives that are correct)
- $F_1$ -Score =  $\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$  (Harmonic mean)

### 🏛️ Solved Exam Flashcard 5: Information Gain vs Gain Ratio vs Gini Index

- **Information Gain (ID3):**  $\text{Gain}(S, A) = H(S) - \sum \frac{|S_v|}{|S|} H(S_v)$ . Biased toward attributes with large numbers of distinct values.
- **Gain Ratio (C4.5):**  $\text{GainRatio}(S, A) = \frac{\text{Gain}(S, A)}{\text{SplitInfo}(S, A)}$ . Normalizes gain by split entropy, penalizing broad multi-way splits.
- **Gini Impurity (CART):**  $\text{Gini}(S) = 1 - \sum p_i^2$ . Strictly binary splits.

### 🏛️ Solved Exam Flashcard 6: Complete Normalization Formulas Reference

- **Min-Max Scaling:**  $v' = \frac{v - \min}{\max - \min} (\text{new\_max} - \text{new\_min}) + \text{new\_min}$
- **Z-Score Standardization:**  $z = \frac{v - \mu}{\sigma}$
- **Modified Z-Score (Outlier-Robust):**  $M_i = \frac{0.6745(x_i - \text{Median})}{\text{MAD}}$  where  $\text{MAD} = \text{Median}(|x_i - \text{Median}|)$
- **Decimal Scaling:**  $v' = \frac{v}{10^j}$  where  $j = \lceil \log_{10}(\max |v|) \rceil$

### 🏛️ Solved Exam Flashcard 7: The 4 Major Clustering Paradigms

| Paradigm             | Representative Algorithms                            | Key Characteristics                                                   |
|----------------------|------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Partitioning</b>  | k-Means, k-Medoids (PAM), CLARA, CLARANS             | Constructs $k$ partitions; iterative relocation; spherical clusters.  |
| <b>Hierarchical</b>  | AGNES (agglomerative), DIANA (divisive), BIRCH, CURE | Dendrogram tree; no $k$ required; irreversible merges.                |
| <b>Density-Based</b> | DBSCAN, OPTICS, DENCLUE                              | Arbitrary non-convex shapes; filters noise outliers automatically.    |
| <b>Grid-Based</b>    | STING, WaveCluster, CLIQUE                           | Quantizes object space into finite grid cells; fast $O(N)$ execution. |

## Solved Flashcard 19: Full Data Preprocessing Pipeline Checklist

- **1. Data Cleaning:** Missing values (impute mean/mode/k-NN), Noise smoothing (bin means, bin boundaries).
- **2. Data Integration:**  $\chi^2$  independence test for nominal data, Pearson  $r$  / Covariance for numeric data.
- **3. Data Reduction:** Dimensionality reduction (PCA, Wavelet, SVD, t-SNE), Numerosity reduction (Histograms, Clustering).
- **4. Data Transformation:** Normalization (Min-Max, Z-score, Decimal scaling), Discretization (ChiMerge, Equi-depth binning).

## Solved Flashcard 20: Data Warehouse Architecture Reference

- **Bottom Tier:** Warehouse Database Server (relational back-end data store).
- **Middle Tier:** OLAP Server (ROLAP relational engines, MOLAP multidimensional servers, or HOLAP hybrid).
- **Top Tier:** Front-End Client Tools (Analysis, Query/Reporting, Data Mining engines).

## Solved Flashcard 21: Full Association Mining Formulas

$$\begin{aligned} \text{Support}(A \implies B) &= \frac{\text{count}(A \cup B)}{N} & \text{Confidence}(A \implies B) &= \frac{\text{count}(A \cup B)}{\text{count}(A)} \\ \text{Lift}(A \implies B) &= \frac{\text{Conf}(A \implies B)}{\text{Support}(B)} & \text{Leverage} &= P(A \cup B) - P(A)P(B) \\ \text{Conviction} &= \frac{1 - \text{Support}(B)}{1 - \text{Conf}(A \implies B)} & \text{Jaccard}(A, B) &= \frac{P(A \cup B)}{P(A) + P(B) - P(A \cup B)} \end{aligned}$$

## Solved Flashcard 22: Classification Decision Criteria

- **Information Gain (ID3):**  $\text{Gain}(S, A) = H(S) - \sum \frac{|S_v|}{|S|} H(S_v)$
- **Gain Ratio (C4.5):**  $\text{GainRatio}(S, A) = \frac{\text{Gain}(S, A)}{\text{SplitInfo}(S, A)}$
- **Gini Impurity (CART):**  $\text{Gini}(S) = 1 - \sum p_i^2$
- **Misclassification Error:**  $\text{Error}(S) = 1 - \max(p_i)$

### 🏛️ Solved Flashcard 23: Complete Clustering Paradigms Summary

| Method        | Core Algorithm                      | Strengths & Limitations                                                                |
|---------------|-------------------------------------|----------------------------------------------------------------------------------------|
| Partitioning  | k-Means, k-Medoids (PAM)            | Fast $O(tkN)$ ; spherical clusters only; sensitive to initialization.                  |
| Hierarchical  | AGNES (bottom-up), DIANA (top-down) | Dendrogram hierarchy; $O(N^2)$ distance matrix; irreversible merges.                   |
| Density-Based | DBSCAN, OPTICS, DENCLUE             | Arbitrary non-convex shapes; filters noise outliers; $O(N \log N)$ with spatial index. |
| Grid-Based    | STING, CLIQUE, WaveCluster          | Quantized spatial grid cells; fast $O(K)$ query speed independent of $N$ .             |

## Master Exam Final Checklists & Complete Solved Cards

### 🏛️ Solved Flashcard 24: KDD vs Data Mining vs Machine Learning

- **KDD**: Complete iterative end-to-end knowledge discovery process (Cleaning, Integration, Selection, Transformation, Mining, Evaluation, Presentation).
- **Data Mining**: The algorithmic pattern discovery phase of KDD.
- **Machine Learning**: Algorithms that learn predictive functions from data.

### 🏛️ Solved Flashcard 25: All Normalization Formulas

- **Min-Max**:  $v' = \frac{v - \min}{\max - \min}(\text{new\_max} - \text{new\_min}) + \text{new\_min}$
- **Z-Score**:  $z = \frac{v - \mu}{\sigma}$
- **Robust MAD**:  $M_i = \frac{0.6745(x_i - \text{Median})}{\text{MAD}}$

### 🏛️ Solved Flashcard 26: Complete Data Warehousing Concepts

- **Inmon**: Top-down 3NF centralized enterprise data warehouse.
- **Kimball**: Bottom-up dimensional data marts with conformed dimensions.
- **Star**: De-normalized, fast OLAP queries.
- **Snowflake**: Normalized dimensions, zero redundancy, multi-join queries.

### 🏛️ Solved Flashcard 27: Association Rule Metrics

$$\text{Support} = \frac{\text{count}(A \cup B)}{N} \quad \text{Confidence} = \frac{\text{count}(A \cup B)}{\text{count}(A)} \quad \text{Lift} = \frac{\text{Conf}}{\text{Support}(B)}$$

### Solved Flashcard 28: Complete Decision Tree Induction Pipeline

Dataset  $S$   $\xrightarrow{\text{Calculate Impurity}}$  Evaluate Splits  $\xrightarrow{\text{Select Max Gain / Min Gini}}$  Partition Tree  $\xrightarrow{\text{Post-Prune Subtrees}}$  Final Model

### Solved Flashcard 29: Complete Hierarchical Clustering Linkage Summary

- **Single Link:**  $d(A, B) = \min d(x, y)$  (Chaining effect, arbitrary shapes)
- **Complete Link:**  $d(A, B) = \max d(x, y)$  (Dense spherical clusters, avoids chaining)
- **Average Link:**  $d(A, B) = \text{mean } d(x, y)$  (Balanced compromise)
- **Ward's Method:** Minimizes total within-cluster variance increase  $\Delta\text{SSE}$

### Solved Flashcard 30: Complete Association Rule Quality Check

$\text{Confidence}(A \implies B) \geq \text{min\_conf}$  AND  $\text{Lift}(A \implies B) > 1.0$  AND  $\chi^2 > 3.841$

### Solved Flashcard 31: OLAP Storage Architecture Summary

- **ROLAP:** Relational backend, Star/Snowflake schemas, high scalability, SQL engine.
- **MOLAP:** Multidimensional array backend, dense matrix offsets, sub-second query latency.
- **HOLAP:** Hybrid engine: relational for detailed leaf records, multidimensional arrays for aggregates.

### Solved Flashcard 32: Outlier Detection Techniques Compared

- **Statistical:**  $3\sigma$  rule, Grubbs' test, Tukey's IQR fences (assumes parametric distribution).
- **Distance-Based:**  $k$ -NN distance thresholding ( $O(N^2)$ ).
- **Density-Based:** Local Outlier Factor (LOF) comparing local reachability densities.
- **Ensemble:** Isolation Forest (iTrees) measuring average path lengths to isolate anomalies in  $O(N \log N)$ .

### 🏛️ Solved Flashcard 33: Complete Model Performance Metrics Reference

| Metric               | Formula                                   | Diagnostic Focus                                   |
|----------------------|-------------------------------------------|----------------------------------------------------|
| Accuracy             | $\frac{TP+TN}{N}$                         | Overall proportion of correct classifications.     |
| Sensitivity (Recall) | $\frac{TP}{TP+FN}$                        | Minimizing False Negatives (medical diagnosis).    |
| Specificity          | $\frac{TN}{TN+FP}$                        | True Negative rate ( $1 - \text{FPR}$ ).           |
| Precision            | $\frac{TP}{TP+FP}$                        | Minimizing False Positives (search ranking, spam). |
| $F_1$ -Score         | $\frac{2 \cdot P \cdot R}{P+R}$           | Harmonic mean for imbalanced targets.              |
| ROC AUC              | $\int_0^1 \text{TPR}(t) d(\text{FPR}(t))$ | Threshold-independent ranking performance.         |

### 🏛️ Solved Flashcard 34: Data Preprocessing Matrix & Transformation Techniques

- **Data Cleaning:** Missing imputation (MICE, k-NN), Noise smoothing (bin means/boundaries).
- **Data Integration:**  $\chi^2$  independence test ( $df = (r - 1)(c - 1)$ ), Pearson correlation  $r_{A,B}$ .
- **Data Transformation:** Min-Max to  $[0, 1]$ , Z-score standardization ( $\mu = 0, \sigma = 1$ ), Decimal scaling.
- **Data Reduction:** PCA ( $EVR \geq 95\%$ ), DWT Wavelets, SVD low-rank approximation, ReliefF feature selection.

### 🏛️ Solved Flashcard 35: Apriori Join & Prune Step Rule

Two frequent  $(k - 1)$ -itemsets  $p, q \in L_{k-1}$  are joined if they share the first  $k - 2$  items ( $p[1] = q[1], \dots, p[k - 2] = q[k - 2]$  and  $p[k - 1] < q[k - 1]$ ). Candidate  $c = p \cup \{q[k - 1]\}$  is pruned if ANY of its  $(k - 1)$  subsets is missing from  $L_{k-1}$ !

### 🏛️ Solved Flashcard 36: DBSCAN Point Categorization Rule

- **Core Point:**  $|N_\epsilon(p)| \geq \text{MinPts}$  (Density center)
- **Border Point:**  $|N_\epsilon(p)| < \text{MinPts}$ , but  $p \in N_\epsilon(q)$  where  $q$  is a core point
- **Noise Point:** Neither core nor border (Isolated anomaly)

### 🏛️ Solved Flashcard 37: Multi-Dimensional Schema Architecture Rule

- **Star Schema:** Fact table surrounded by flat de-normalized dimension tables.
- **Snowflake Schema:** Fact table surrounded by normalized dimension hierarchy tables.
- **Fact Constellation:** Multiple fact tables sharing conformed dimensions.

### 🏛️ Solved Flashcard 38: Attribute Subset Selection Techniques

- **Stepwise Forward Selection:** Starts with  $\emptyset$ , greedily adds best attribute.
- **Stepwise Backward Elimination:** Starts with full feature set, greedily removes worst attribute.
- **Decision Tree Induction:** Attributes selected as split nodes represent minimal descriptive subset.

### 🏛️ Solved Flashcard 39: Classification Performance Metrics Summary

$$\text{FPR} = \frac{FP}{TN + FP} = 1 - \text{Specificity} \quad \text{FNR} = \frac{FN}{TP + FN} = 1 - \text{Sensitivity}$$

### 🏛️ Solved Flashcard 40: Density-Based Reachability Axioms

$$p \xrightarrow{\text{direct}} q \implies q \in N_\epsilon(p) \wedge |N_\epsilon(p)| \geq \text{MinPts}$$

Density-Reachable: Chain of Core Points

Density-Connected: Common Core Ancestor  $o$

### 🏛️ Solved Flashcard 41: Data Cube Computation Strategies

- **BUC (Bottom-Up Computation):** Explores cuboid tree bottom-up with Apriori pruning; ideal for sparse iceberg cubes.
- **MultiWay Array Aggregation:** Computes full dense cubes in single pass over multidimensional RAM chunks.
- **Star-Cubing:** Combines star-trees and shared prefix paths for iceberg cubing.

### 🏛️ Solved Flashcard 42: Distance Metrics Axioms & Semi-Metrics

$$\text{Non-negativity: } d(x, y) \geq 0 \quad \text{Identity: } d(x, y) = 0 \iff x = y$$

$$\text{Symmetry: } d(x, y) = d(y, x) \quad \text{Triangle Inequality: } d(x, z) \leq d(x, y) + d(y, z)$$

### 🏛️ Solved Flashcard 43: SVD vs PCA Mathematical Equivalences

$$\text{Centering: } \mathbf{X}_c = \mathbf{X} - \mathbf{1}\mu^T \implies \mathbf{X}_c = \mathbf{U}\Sigma\mathbf{V}^T$$

$$\text{Covariance: } \mathbf{S} = \frac{1}{n-1} \mathbf{X}_c^T \mathbf{X}_c = \frac{1}{n-1} \mathbf{V}\Sigma^2\mathbf{V}^T \implies \lambda_i = \frac{\sigma_i^2}{n-1}, \mathbf{e}_i = \mathbf{v}_i$$

### 🏛️ Solved Flashcard 44: Complete Hierarchical Linkage Metrics

- **Single Link (Nearest Neighbor):**  $d(A, B) = \min_{x \in A, y \in B} d(x, y)$
- **Complete Link (Farthest Neighbor):**  $d(A, B) = \max_{x \in A, y \in B} d(x, y)$
- **Average Link (Group Average):**  $d(A, B) = \frac{1}{|A||B|} \sum_{x \in A} \sum_{y \in B} d(x, y)$
- **Centroid Link:**  $d(A, B) = \|\mu_A - \mu_B\|_2$

### **Solved Flashcard 45: Complete Master Review Strategy**

Master the 4 Core Pillars: 1. Proximity & Five-number summary, 2. Star/Snowflake & OLAP lattices, 3. Apriori/FP-Tree association rules, 4. Decision trees & DBSCAN density clustering!